

SOFTWARE REQUIREMENTS SPECIFICATION

AI-Powered Multi-Tenant IVR Platform

Conforming to IEEE Std 830-1998 Recommended Practice for Software Requirements Specifications

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4. Introduction

4.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the AI-Powered Multi-Tenant IVR Platform. It is intended to serve as the authoritative reference for the development team, project supervisors, and evaluators throughout design, implementation, and testing. This document conforms to the structure recommended by IEEE Std 830-1998.

4.2 Scope

The AI-Powered Multi-Tenant IVR Platform is a cloud-based, multi-tenant B2B SaaS system that allows independent companies (tenants) to design, deploy, and operate their own Interactive Voice Response (IVR) systems on shared platform infrastructure. Each tenant manages its own organizational structure (departments, employees, extensions), builds call-handling logic through a visual flow builder, and receives, routes, records, and reports on inbound telephone calls through an integrated Asterisk PBX telephony layer.

The platform additionally provides an AI assistant that helps Tenant Administrators generate IVR flows, generate voice prompts, receive workflow-improvement recommendations, and analyze reports. The AI assistant supports administrative configuration only; it does not participate in answering live customer calls, which remain fully handled by the deterministic IVR execution engine.

This document covers requirements for the web application (React.js frontend, Java Servlet/JDBC backend), the telephony integration layer (Asterisk/SIP/PJSIP), and the AI integration layer. Database schema design, entity-relationship diagrams, and source code are explicitly out of scope for this document.

4.3 Definitions

Term	Definition
Tenant	A company or organization that has registered on the platform and operates its own isolated IVR configuration.
Flow	A named IVR workflow designed by a tenant, composed of connected nodes.
Flow Version	An immutable, published snapshot of a Flow's node graph that can be deployed.
Node	A single configurable step within a Flow (e.g., Greeting, Menu, Queue).
DID	Direct Inward Dial number; the public phone number that routes to a tenant's deployed Flow.
CDR	Call Detail Record; the finalized historical record of a completed call.
DTMF	Dual-Tone Multi-Frequency; the signaling used for touch-tone keypad input.
Extension	A SIP endpoint assigned to an employee for receiving/making calls.
Queue	A holding mechanism that routes waiting callers to available agents.

4.4 Acronyms

Acronym	Meaning
SRS	Software Requirements Specification
IVR	Interactive Voice Response
SaaS	Software as a Service
PBX	Private Branch Exchange
SIP	Session Initiation Protocol
PJSIP	A SIP stack/module used by Asterisk for endpoint management
RBAC	Role-Based Access Control
API	Application Programming Interface
TTS	Text-to-Speech
CDR	Call Detail Record
DID	Direct Inward Dial
AMI/ARI	Asterisk Manager Interface / Asterisk REST Interface
DAO	Data Access Object
MVC	Model-View-Controller
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case

4.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Asterisk PBX Documentation — Asterisk Project.
- PJSIP / res_pjsip Configuration Reference — Asterisk Project.
- Project Business & Architecture Analysis (prior deliverable in this graduation project).

4.6 Document Overview

Section 5 describes the product at a high level. Section 6 lists detailed functional requirements. Section 7 lists non-functional requirements. Section 8 defines user roles. Section 9 lists cross-cutting business rules. Section 10 defines external interfaces. Section 11 provides detailed use cases. Sections 12-15 cover acceptance criteria, future enhancements, risks, and a glossary.

5. Overall Description

5.1 Product Perspective

The platform is a new, self-contained SaaS product. It is not an add-on to an existing system, though it integrates with external systems in two respects: (1) it uses an external LLM API for AI-assisted features, and (2) it relies on Asterisk PBX as its telephony engine, integrated via SIP/PJSIP and Asterisk's control interfaces. The web application (React.js frontend and Java Servlet/JDBC backend) is the system of record for all tenant configuration; Asterisk executes the real-time call handling instructed by the platform's IVR execution engine.

5.2 Product Functions

At a high level, the platform:

- Allows companies to register and manage an isolated tenant workspace.
- Allows tenants to manage departments, employees, and SIP extensions.
- Allows tenants to visually design, validate, version, and deploy IVR flows.
- Executes deployed flows in real time to handle inbound calls, including menus, queues, transfers, recording, and voicemail.
- Generates reports and analytics on call and flow performance.
- Provides AI assistance for flow generation, prompt generation, workflow improvement suggestions, and report analysis.

5.3 User Classes and Characteristics

User Class	Characteristics
Super Admin	Anthropic-style platform operator; technical, manages tenant lifecycle and platform health.
Tenant Admin	Business administrator of a tenant company; non-technical to moderately technical; primary user of the IVR builder and configuration screens.
Manager	Departmental manager within a tenant; views reports and manages employees/queues within their scope.
Agent (Employee)	Handles transferred calls and voicemail; minimal interaction with the web app beyond status and voicemail retrieval.
Auditor	Read-only access to reports, audit logs, and compliance data within a tenant.
Caller	External customer interacting with the IVR by voice/DTMF only; has no platform account.

5.4 Operating Environment

- Server-side: Apache Tomcat running Java Servlets on Linux, containerized with Docker.
- Database: PostgreSQL, deployed as a containerized service.
- Telephony: Asterisk PBX with PJSIP channel driver, containerized or deployed on dedicated telephony-optimized infrastructure.
- Client: Modern evergreen web browsers (Chrome, Firefox, Edge, Safari) running the React.js single-page application.
- AI Integration: Outbound HTTPS calls to an external LLM API provider.

5.5 Design Constraints

- Backend must be implemented using Core Java, Java Servlets, JDBC, and the DAO pattern (no full-stack framework such as Spring), per project scope.
- All components must be deployable via Docker containers.
- Telephony must be built on Asterisk PBX using SIP/PJSIP; no alternative telephony engine is in scope.
- The relational database must be PostgreSQL.
- The AI assistant must not be used to answer live customer calls; its role is strictly administrative/assistive.

5.6 Assumptions

- Tenants have access to their own PSTN connectivity or will use trunking arranged through the platform's SIP trunk provider(s).
- Tenants have basic familiarity with web applications; no telephony engineering expertise is assumed for day-to-day IVR configuration.
- An external LLM API will remain available and reachable via HTTPS from the platform's backend.
- Initial deployment targets a moderate number of tenants and concurrent calls consistent with a graduation-project-scale production pilot, not global hyperscale volume.

5.7 Dependencies

- Availability and correctness of the external LLM API for all AI-assisted features.
- Availability of SIP trunk/PSTN connectivity for each tenant's inbound and outbound calling.
- Correct configuration of Asterisk PJSIP endpoints matching platform-provisioned SIP extensions.
- Docker runtime environment for deployment of all system components.

6. Functional Requirements

This section enumerates all functional requirements (FR) of the platform. Requirements are grouped informally by domain but numbered sequentially. Each requirement includes its priority, actors, preconditions, postconditions, and governing business rules.

FR-001 — Tenant Registration

Description: The platform shall allow a new business to register as a tenant by providing company information (name, industry type, contact details, timezone) and creating an initial Tenant Admin account.

Priority: High **Actors:** Prospective Tenant (Company Representative)

Preconditions: The registrant has a valid business email address and has not already registered the same company.

Postconditions: A new Tenant record is created in a Pending status; a Tenant Admin User account is created and linked to the tenant.

Business Rules:

- A tenant cannot access platform features until approved (see FR-002), except for viewing its own pending status.
- Each tenant must have a globally unique identifier and a unique primary contact email.
- Registration must capture the tenant's industry type, which may later influence default IVR templates.

FR-002 — Tenant Approval and Lifecycle Management

Description: The platform shall allow a Super Admin to review, approve, reject, suspend, or reactivate tenant accounts.

Priority: High **Actors:** Super Admin

Preconditions: A tenant exists in Pending, Active, or Suspended status.

Postconditions: The tenant's status is updated accordingly; the Tenant Admin is notified of the status change.

Business Rules:

- Only a Super Admin may change a tenant's lifecycle status.
- A suspended tenant's telephony services (inbound call routing) must be disabled while suspended.
- Rejected or offboarded tenants must have a defined data retention/deletion policy applied.

FR-003 — User Authentication

Description: The platform shall allow registered users to authenticate using a username/email and password, scoped to their tenant, and shall support session expiry and logout.

Priority: High **Actors:** Super Admin, Tenant Admin, Manager, Agent, Auditor

Preconditions: The user has an active account within an active tenant.

Postconditions: On success, an authenticated session is established; on failure, access is denied and the attempt is logged.

Business Rules:

- Passwords must meet a configurable minimum complexity policy.
- Accounts must be locked after a configurable number of consecutive failed login attempts.
- A user may only authenticate into the tenant(s) they belong to; cross-tenant login is prohibited.
- Sessions must expire after a configurable period of inactivity.

FR-004 — Role-Based Access Control (RBAC)

Description: The platform shall enforce access to features and data based on the authenticated user's assigned role(s) (Super Admin, Tenant Admin, Manager, Agent, Auditor).

Priority: High **Actors:** Super Admin, Tenant Admin

Preconditions: Roles and permissions are defined in the system.

Postconditions: The user is granted or denied access to a requested function/resource based on their role's permission set.

Business Rules:

- Every protected action must check both the user's role and tenant scope before executing.
- A Tenant Admin may define additional custom roles scoped to their own tenant only.
- Permission checks must be enforced server-side, not solely in the UI.

FR-005 — Department Management

Description: The platform shall allow a Tenant Admin to create, edit, and deactivate departments within their tenant, including optional department-specific business hours.

Priority: Medium **Actors:** Tenant Admin

Preconditions: The tenant is active.

Postconditions: The department is available for assignment to employees, queues, and IVR nodes.

Business Rules:

- A department must belong to exactly one tenant.
- A department cannot be deleted if it has active employees or queues assigned; it must first be deactivated.

FR-006 — Employee Management

Description: The platform shall allow a Tenant Admin or Manager to create, edit, and deactivate employee records, assign employees to departments, and optionally link an employee to a system User account.

Priority: High **Actors:** Tenant Admin, Manager

Preconditions: The tenant is active; the target department exists (if assigned).

Postconditions: The employee record is available for extension assignment and queue membership.

Business Rules:

- An employee may belong to more than one department or queue.
- Deactivating an employee must automatically remove them from all active queue memberships.

FR-007 — SIP Extension Provisioning

Description: The platform shall allow a Tenant Admin to provision a SIP extension, generate its credentials, and assign it to an employee.

Priority: High **Actors:** Tenant Admin

Preconditions: The tenant is active; the employee exists and does not already hold an active extension of the same type.

Postconditions: A SIP extension is created and registered as available for the Asterisk PJSIP endpoint configuration.

Business Rules:

- Extension numbers must be unique within a tenant.
- SIP credentials must be generated with sufficient entropy and never displayed in plaintext after initial creation.
- Deactivating an employee must automatically deactivate their associated extension(s).

FR-008 — Call Queue Management

Description: The platform shall allow a Tenant Admin to create and configure call queues, including routing strategy (round robin, longest idle, skills-based), maximum wait time, and overflow behavior.

Priority: High **Actors:** Tenant Admin

Preconditions: The tenant is active.

Postconditions: The queue is available for selection within IVR Queue nodes and for employee membership assignment.

Business Rules:

- A queue must define an overflow action (e.g., route to voicemail) for when the max wait time is exceeded.
- A queue with zero active member employees must still accept callers but should trigger an operational alert.

FR-009 — Visual IVR Flow Builder

Description: The platform shall provide a drag-and-drop visual editor allowing a Tenant Admin to create and edit IVR flows as a graph of connected nodes.

Priority: High **Actors:** Tenant Admin

Preconditions: The tenant is active.

Postconditions: A Flow (in Draft status) reflecting the designed graph is saved.

Business Rules:

- Every flow must have exactly one designated entry node.
- Edits to a published flow must be made on a new draft version, never in place on the published version.
- The builder must support all defined node types (Greeting, Playback, Menu, DTMF Input, Queue, Agent Transfer, Voicemail, Recording, API Request, Database Lookup, Business Hours, Holiday Check, Hangup).

FR-010 — IVR Node Configuration

Description: The platform shall allow each node in a flow to be individually configured according to its type (e.g., a Menu node configures DTMF options and destinations; an API Request node configures an endpoint and mapping).

Priority: High **Actors:** Tenant Admin

Preconditions: A flow draft exists and the node has been placed on the canvas.

Postconditions: The node's configuration is persisted as part of the flow draft.

Business Rules:

- Each node type must define its own required configuration fields, and the builder must prevent saving a node missing required fields.
- API Request and Database Lookup nodes must support configurable timeout and fallback-path behavior in case of failure.

FR-011 — Flow Validation

Description: The platform shall validate a flow draft before allowing it to be published, checking for structural issues such as orphan nodes, missing required connections, and unreachable end states.

Priority: High **Actors:** Tenant Admin, System

Preconditions: A flow draft exists.

Postconditions: The flow is marked as Valid or Invalid, with a list of validation errors if applicable.

Business Rules:

- A flow with unresolved validation errors cannot be published or deployed.
- Every path through the flow must terminate in a defined end state (Hangup, Queue, Agent Transfer, or Voicemail).

FR-012 — Flow Versioning

Description: The platform shall maintain immutable, numbered versions of a flow each time it is published, and shall allow a Tenant Admin to view version history and roll back to a previous version.

Priority: High **Actors:** Tenant Admin

Preconditions: At least one version of the flow has been published.

Postconditions: The selected version becomes the active/deployed version for its assigned DID(s).

Business Rules:

- A published Flow Version can never be edited; changes require creating a new draft and publishing a new version.
- Rolling back must not delete newer versions; it only changes which version is currently deployed.

FR-013 — Flow Deployment

Description: The platform shall allow a Tenant Admin to assign a published Flow Version to one or more DIDs (phone numbers), making it live for inbound calls.

Priority: High **Actors:** Tenant Admin

Preconditions: A Flow Version exists in Published/Valid status; the target DID belongs to the same tenant.

Postconditions: Inbound calls to the DID are routed to the newly deployed Flow Version.

Business Rules:

- A DID may be mapped to only one active Flow Version at a time.
- Deployment changes must take effect for new calls without requiring a platform restart.

FR-014 — Voice Prompt Upload

Description: The platform shall allow a Tenant Admin to upload audio files as voice prompts, tagging them with a name and language/locale for use in Playback and Greeting nodes.

Priority: Medium **Actors:** Tenant Admin

Preconditions: The tenant is active.

Postconditions: The uploaded prompt is available for selection within the IVR builder.

Business Rules:

- Uploaded audio must conform to a supported format/codecs compatible with Asterisk playback.
- A prompt in use by at least one published flow cannot be deleted; it may only be archived.

FR-015 — AI-Assisted Voice Prompt Generation (Text-to-Speech)

Description: The platform shall allow a Tenant Admin to generate a voice prompt from typed text using an external AI/TTS service, and preview the result before saving it as a Voice Prompt asset.

Priority: Medium **Actors:** Tenant Admin, AI Service (external)

Preconditions: The tenant has AI features enabled under its plan.

Postconditions: A new Voice Prompt asset is created from the generated audio, marked with its generation source.

Business Rules:

- The admin must explicitly accept the generated audio before it becomes a usable prompt.
- Generation requests must be metered against the tenant's usage allowance, if applicable.

FR-016 — Inbound Call Reception and Tenant/Flow Resolution

Description: The platform shall receive an inbound call via Asterisk, resolve the dialed DID to its owning tenant and currently deployed Flow Version, and begin flow execution.

Priority: High **Actors:** System (Telephony Layer), Caller

Preconditions: The DID is active and mapped to a deployed Flow Version.

Postconditions: A Call Session record is created and the flow begins execution at its entry node.

Business Rules:

- If a DID has no deployed flow, the call must be routed to a configurable default fallback (e.g., generic greeting and hangup) rather than failing silently.
- Tenant resolution must occur before any tenant-scoped data is accessed for the call.

FR-017 — IVR Runtime Execution Engine

Description: The platform shall execute a deployed Flow Version node-by-node for the duration of a call, evaluating each node's configured behavior and following connections based on outcomes.

Priority: High **Actors:** System (Execution Engine)

Preconditions: A Call Session has been created and an entry node identified.

Postconditions: The call reaches a terminal node (Hangup, Queue hand-off, Agent Transfer, or Voicemail) and the session is marked accordingly.

Business Rules:

- Execution state must be tracked per call session so concurrent calls do not interfere with one another.
- If a node fails (e.g., an API Request timeout), the engine must follow the node's configured fallback path rather than terminating the call abruptly.
- Every node visited must emit an Analytics Event for later reporting.

FR-018 — DTMF Input Handling

Description: The platform shall capture caller DTMF input (single digit or multi-digit) as specified by Menu and DTMF Input nodes, and route the call according to the configured mapping.

Priority: High **Actors:** Caller, System

Preconditions: The flow execution has reached a Menu or DTMF Input node.

Postconditions: The call is routed along the connection matching the entered digit(s), or the 'no input'/'invalid input' path if applicable.

Business Rules:

- Every Menu node must define a timeout and an invalid-input behavior (e.g., repeat the menu up to N times, then fall back).
- Digit collection must respect a configurable maximum number of digits and inter-digit timeout.

FR-019 — Call Queueing and Agent Routing

Description: The platform shall place a caller into the configured Call Queue when execution reaches a Queue node, and route the caller to an available agent according to the queue's routing strategy.

Priority: High **Actors:** Caller, System, Employee (Agent)

Preconditions: The caller has reached a Queue node; the queue exists and is active.

Postconditions: The caller is connected to an agent, or the queue's overflow action is triggered if wait time is exceeded.

Business Rules:

- Agents must be selectable only if their status is Available.
- Wait time must be tracked continuously and compared against the queue's configured maximum.

FR-020 — Call Transfer

Description: The platform shall support transferring an active call to a specific agent extension (Agent Transfer node) or to a queue, either as a blind or attended transfer.

Priority: High **Actors:** System, Employee (Agent)

Preconditions: The target extension or queue exists and is active.

Postconditions: The call is connected to the target; the Call Session records the transfer event.

Business Rules:

- If the target extension is unreachable, the flow must follow a configured fallback path (e.g., voicemail).

FR-021 — Call Recording

Description: The platform shall record calls when a flow reaches a Recording node or when tenant-wide recording is enabled, store the resulting audio, and associate it with the Call Session.

Priority: High **Actors:** System

Preconditions: Recording is enabled for the tenant, department, or specific flow node.

Postconditions: A Recording entity is created, linked to the Call Session, and made available for authorized playback.

Business Rules:

- Where required by tenant configuration, a consent announcement must be played and a Consent Record captured before recording begins.
- Recordings must be retained according to the tenant's configured retention policy and purged automatically thereafter.
- Access to recordings must be restricted to authorized roles and logged.

FR-022 — Voicemail Capture

Description: The platform shall allow a caller to leave a voicemail message when routed to a Voicemail node, storing the audio and notifying the assigned employee or department.

Priority: Medium **Actors:** Caller, System, Employee

Preconditions: Flow execution has reached a Voicemail node.

Postconditions: A Voicemail record is created with status 'New' and the assigned recipient is notified.

Business Rules:

- A maximum recording duration must be enforced for voicemail messages.
- Voicemail must be retrievable only by the assigned employee, their department, or an authorized admin.

FR-023 — Call Detail Record (CDR) Generation

Description: The platform shall generate a finalized Call Detail Record at the end of every call, capturing timestamps, duration, disposition, and the path taken through the flow.

Priority: High **Actors:** System

Preconditions: A Call Session has ended (hangup).

Postconditions: An immutable CDR is persisted and made available to the Reporting module.

Business Rules:

- CDRs must never be edited after creation; corrections require a compensating record, not an in-place update.
- Every CDR must be scoped to exactly one tenant.

FR-024 — Reporting Dashboard

Description: The platform shall provide dashboards summarizing call volume, queue performance, agent performance, and IVR flow usage, filterable by date range and department.

Priority: High **Actors:** Tenant Admin, Manager, Auditor

Preconditions: CDR and analytics data exist for the requested period.

Postconditions: The requested report/dashboard is rendered or exported (CSV/PDF).

Business Rules:

- A user may only view reports scoped to their own tenant and, where applicable, their own department.
- Exported reports must not include data beyond the requesting user's authorized scope.

FR-025 — IVR Flow Analytics

Description: The platform shall analyze Analytics Events to compute per-flow metrics such as node-level drop-off rate, average time-to-completion, and most common paths.

Priority: Medium **Actors:** Tenant Admin, System

Preconditions: The flow has processed at least one completed call.

Postconditions: Flow analytics are available for display in the builder or dashboard.

Business Rules:

- Analytics must be computed per Flow Version so comparisons between versions (e.g., before/after an AI-suggested change) are possible.

FR-026 — AI-Assisted IVR Flow Generation

Description: The platform shall allow a Tenant Admin to describe a desired IVR behavior in natural language and receive an AI-generated draft Flow Version for review.

Priority: High **Actors:** Tenant Admin, AI Service (external)

Preconditions: The tenant has AI features enabled.

Postconditions: A new draft Flow (not published) is created, reflecting the AI-generated node graph, and is subject to full validation (FR-011) before it can be published.

Business Rules:

- AI-generated flows must always be created as drafts; they may never be published automatically without explicit Tenant Admin review and approval.
- The AI request and its output must be logged as an AI Request/Generation Job for traceability.

FR-027 — AI Workflow Improvement Recommendations

Description: The platform shall allow a Tenant Admin to request AI-generated suggestions for improving an existing flow (e.g., reducing steps, adding fallback paths), based on the flow's structure and its analytics.

Priority: Medium **Actors:** Tenant Admin, AI Service (external)

Preconditions: A published flow with analytics data exists.

Postconditions: A set of suggested changes is presented to the admin; if accepted, a new draft version incorporating the changes is created.

Business Rules:

- Suggestions must never be applied automatically to a live/published flow.
- The admin may accept, reject, or partially apply suggestions.

FR-028 — AI Report Analysis and Summarization

Description: The platform shall allow a Tenant Admin to request an AI-generated plain-language summary of report/analytics data for a given period, including notable anomalies.

Priority: Low **Actors:** Tenant Admin, AI Service (external)

Preconditions: Report data exists for the requested period.

Postconditions: A natural-language summary is displayed alongside the underlying report.

Business Rules:

- The AI summary must be clearly labeled as AI-generated and must not replace the underlying numeric report.
- Only aggregated, tenant-scoped data may be sent to the external AI service — no raw recordings or personally identifying caller data.

FR-029 — Audit Logging

Description: The platform shall record an audit log entry for every significant configuration change (flow edits, permission changes, tenant setting changes, user role changes).

Priority: High **Actors:** System

Preconditions: A user performs an auditable action.

Postconditions: An Audit Log Entry is persisted, capturing the actor, action, target, and timestamp.

Business Rules:

- Audit logs must be append-only and tamper-evident.
- Audit logs must be retained for a minimum period defined by the tenant's compliance requirements.

FR-030 — Business Hours and Holiday Configuration

Description: The platform shall allow a Tenant Admin to configure business hours and holiday calendars at the tenant or department level, for use in Business Hours and Holiday Check nodes.

Priority: Medium **Actors:** Tenant Admin

Preconditions: The tenant is active.

Postconditions: The configured schedule is available for evaluation by flow nodes at call time.

Business Rules:

- A department-level business-hours override, where defined, takes precedence over the tenant-level default.
- Time evaluation must always use the tenant's configured timezone, not the server's local timezone.

FR-031 — Multi-Tenant Data Isolation Enforcement

Description: The platform shall ensure that every data access operation is scoped to the requesting user's tenant, preventing any cross-tenant read or write access.

Priority: High **Actors:** System

Preconditions: A user or system process requests access to a resource.

Postconditions: Access is granted only if the resource belongs to the requester's tenant (or the requester is a Super Admin acting in a platform-administration capacity).

Business Rules:

- Tenant scoping must be enforced at the data access layer (DAO level), not solely in UI logic or controller code.
- Any request attempting to access another tenant's resource must be denied and logged as a security event.

FR-032 — Platform Administration Console

Description: The platform shall provide a Super Admin console for overseeing all tenants, monitoring platform health, and managing platform-wide configuration.

Priority: Medium **Actors:** Super Admin

Preconditions: The user is authenticated with the Super Admin role.

Postconditions: The Super Admin can view/manage tenant status, platform usage summaries, and global settings.

Business Rules:

- Super Admin actions affecting a specific tenant must still be recorded in that tenant's audit trail where applicable.

7. Non-Functional Requirements

7.1 Performance

- NFR-P1: The IVR execution engine shall process each node transition (excluding external API/DB lookups) within 300 ms under normal load.
- NFR-P2: The web application's admin-facing pages shall return within 2 seconds under normal load for 95% of requests.
- NFR-P3: The platform shall support at least 100 concurrent active calls per telephony node in the initial deployment target, with a defined path to horizontal scaling.

7.2 Availability

- NFR-A1: The call-handling path (DID reception through flow execution) shall target 99.9% monthly uptime.
- NFR-A2: The administrative web application shall target 99.5% monthly uptime.
- NFR-A3: Planned maintenance windows shall be scheduled and communicated to tenants in advance.

7.3 Reliability

- NFR-R1: A failure in a single node's execution (e.g., an API Request timeout) shall not terminate the call; the engine shall follow the node's configured fallback path.
- NFR-R2: CDRs shall be generated for 100% of calls that reach the execution engine, including calls that end abnormally.
- NFR-R3: The system shall recover call-handling capability automatically after a transient failure of a dependent service (database, AI service) without requiring manual restart of the telephony layer.

7.4 Scalability

- NFR-S1: The telephony layer and the web/API layer shall be independently scalable (separate deployment units).
- NFR-S2: The data model and query design shall support tenant-scoped partitioning strategies to prevent one high-volume tenant from degrading performance for others.
- NFR-S3: The AI integration shall be implemented as an asynchronous job so AI request volume does not block core platform operations.

7.5 Security

- NFR-SEC1: All data in transit shall be encrypted using TLS; all sensitive data at rest (credentials, recordings) shall be encrypted.
- NFR-SEC2: Tenant data isolation shall be enforced at the data access layer for every query and command.
- NFR-SEC3: SIP extension credentials shall be generated with sufficient entropy and shall never be transmitted or displayed in plaintext after initial provisioning.

- NFR-SEC4: The platform shall implement rate limiting and anomaly detection to mitigate toll fraud and call-flooding attacks.
- NFR-SEC5: All authentication and authorization failures shall be logged as security events.

7.6 Maintainability

- NFR-M1: The backend codebase shall be organized into clearly separated packages per business module (identity, tenant, organization, telephony, ivr.design, ivr.execution, reporting, ai) to compensate for the absence of a framework-enforced layering.
- NFR-M2: New IVR node types shall be addable without modifying the execution logic of existing node types.

7.7 Usability

- NFR-U1: The visual IVR builder shall be operable by a non-technical Tenant Admin without requiring code or scripting knowledge.
- NFR-U2: The platform shall provide inline validation feedback (e.g., missing node configuration) at the point of error, not only at publish time.

7.8 Portability

- NFR-PO1: All platform components (Tomcat/Servlets, PostgreSQL, Asterisk) shall be deployable via Docker containers on any Docker-compatible host environment.

7.9 Compatibility

- NFR-C1: The React.js frontend shall be compatible with current major versions of Chrome, Firefox, Edge, and Safari.
- NFR-C2: The telephony layer shall be compatible with standard SIP trunk providers supporting PJSIP registration.

7.10 Backup

- NFR-B1: The PostgreSQL database shall be backed up on a scheduled basis with a defined retention period.
- NFR-B2: Call recordings and voicemail audio shall be included in the backup strategy, subject to each tenant's retention policy.

7.11 Logging

- NFR-L1: All configuration changes shall be logged via the Audit Logging function (FR-029).
- NFR-L2: All system errors and exceptions in the execution engine shall be logged with sufficient context (call session ID, node ID) to support troubleshooting.

7.12 Monitoring

- NFR-MO1: The platform shall expose health-check endpoints for the web application and the telephony integration layer.

- NFR-MO2: The platform shall support alerting on key operational thresholds (e.g., queue SLA breaches, elevated call failure rate).

8. User Roles

Super Admin

Operates the platform itself. Approves/suspends tenants, oversees platform health, manages global configuration. Not scoped to any single tenant's business data by default.

Tenant Admin

The primary administrative user for a company. Manages departments, employees, extensions, queues, IVR flows, voice prompts, and views all reports within their own tenant. Uses the AI assistant.

Manager

A scoped administrator, typically limited to a department. Manages employees and queues within their department and views reports for that scope.

Agent (Employee)

Receives transferred calls and voicemail. Interacts primarily with a softphone/device; limited web app access to status and voicemail.

Auditor

Read-only role with access to reports, CDRs, and audit logs for compliance review within a tenant.

Caller

An external party who dials a tenant's DID. Has no platform account and interacts only through voice and DTMF input.

9. Business Rules

- BR-01: Every business entity except platform-global entities (e.g., Super Admin accounts) must belong to exactly one Tenant.
- BR-02: A Flow Version, once published, is immutable; changes always occur through a new draft and a new version.
- BR-03: A DID may be mapped to only one actively deployed Flow Version at a time.
- BR-04: AI-generated or AI-suggested content (flows, prompts, recommendations) must always require explicit human review and approval before taking effect in a live/published state.
- BR-05: Recording of a call requires either tenant-wide policy enablement or an explicit Recording node in the flow, and must respect the tenant's configured consent requirements.
- BR-06: An employee's SIP extension and queue memberships are automatically deactivated when the employee is deactivated.
- BR-07: A department cannot be deleted while it has active employees or queues; it must be archived instead.
- BR-08: Every completed call must produce exactly one immutable CDR.
- BR-09: A suspended tenant's inbound call routing is disabled, but in-progress calls at the time of suspension are allowed to complete.
- BR-10: All time-based flow logic (Business Hours, Holiday Check) must evaluate against the tenant's configured timezone.

10. External Interfaces

10.1 User Interface

A responsive React.js single-page application providing: authentication screens, tenant/organization management screens, the visual IVR flow builder (graph canvas with node palette and configuration panels), voice prompt management, reporting dashboards, and the AI assistant panel. The UI must reflect role-based access, hiding or disabling functions unavailable to the current user's role.

10.2 Database Interface

The Java backend accesses PostgreSQL exclusively through JDBC using the DAO pattern; no direct database access is exposed to the frontend. Connection pooling and prepared statements are required to prevent SQL injection and manage resource usage under concurrent load.

10.3 Asterisk Interface

The platform integrates with Asterisk PBX for call signaling and media handling via SIP/PJSIP endpoints and Asterisk's control interface (AMI/ARI). This interface is responsible for: registering/managing SIP extensions and trunks, receiving call events (new call, DTMF, hangup), issuing call-control commands (answer, playback, transfer, record, hangup) as directed by the IVR execution engine, and reporting call state changes back to the platform in real time.

10.4 AI Interface

The platform integrates with an external LLM API over HTTPS for: IVR flow generation from natural language, voice prompt text generation/TTS requests, workflow improvement suggestions, and report summarization. Requests are scoped to a tenant's context and, per NFR-SEC and FR-028's business rules, must exclude raw recordings and personally identifying caller data — only aggregated or configuration-level data is sent.

10.5 REST APIs

The backend exposes REST APIs (consumed by the React.js frontend and, where applicable, tenant-side integrations) covering: authentication and session management, tenant and organization management, IVR flow design/versioning/deployment, voice prompt management, queue and call-session status, reporting/analytics retrieval, and AI assistant requests. All APIs enforce authentication, RBAC, and tenant-scoping on every call.

11. Use Cases

This section presents detailed use cases for the platform's major features, complementing the functional requirements in Section 6.

UC-01 — Register a New Tenant

Actors: Prospective Tenant (Company Representative), Super Admin

Preconditions: The registrant has a valid business email not already associated with an existing tenant.

Main Flow:

1. The registrant navigates to the registration page and submits company information and desired admin credentials.
2. The system validates the submitted data (format, uniqueness of email/company).
3. The system creates a Tenant record in Pending status and a Tenant Admin User account.
4. The system sends a confirmation notification to the registrant and an approval request to the Super Admin.
5. The Super Admin reviews and approves the tenant.
6. The system activates the tenant and notifies the Tenant Admin that the account is ready to use.

Alternative Flow:

- 3a. If the company or email already exists, the system rejects the submission and displays an error.
- 5a. If the Super Admin rejects the registration, the tenant is marked Rejected and the registrant is notified with a reason.

Exceptions:

- If the notification service is unavailable, the tenant is still created in Pending status and the notification is retried asynchronously.

Postconditions: A tenant exists in Active status (success path) or Rejected status (alternative path), with the registrant notified either way.

UC-02 — Build and Publish an IVR Flow

Actors: Tenant Admin

Preconditions: The Tenant Admin is authenticated; the tenant is Active.

Main Flow:

7. The Tenant Admin creates a new IVR Flow and opens the visual builder.
8. The Tenant Admin drags nodes onto the canvas and connects them to define the call path.
9. The Tenant Admin configures each node's settings (e.g., prompt selection, menu options, queue target).
10. The Tenant Admin saves the flow as a draft.
11. The Tenant Admin requests validation of the draft.

12. The system validates the graph and reports success.

13. The Tenant Admin publishes the flow, creating a new immutable Flow Version.

14. The Tenant Admin assigns the published version to a DID.

Alternative Flow:

- 6a. If validation fails, the system lists the specific errors (e.g., orphan node, missing fallback) and the flow remains in Draft status until corrected.

Exceptions:

- If the Tenant Admin's session expires mid-edit, unsaved changes are lost and the admin must re-authenticate and resume from the last saved draft.

Postconditions: A validated Flow Version is published and live for inbound calls on the assigned DID.

UC-03 — Generate an IVR Flow Using AI

Actors: Tenant Admin, AI Service (external)

Preconditions: The Tenant Admin is authenticated; the tenant has AI features enabled.

Main Flow:

15. The Tenant Admin opens the AI Assistant panel and describes the desired IVR behavior in natural language.

16. The system sends the request to the external AI service along with relevant tenant context (e.g., industry type, available node types).

17. The AI service returns a proposed flow structure.

18. The system creates a new draft Flow Version reflecting the AI's proposal and opens it in the visual builder for review.

19. The Tenant Admin reviews, edits as needed, validates, and publishes the flow (continuing as in UC-02, steps 5-8).

Alternative Flow:

- 3a. If the AI service returns an incomplete or invalid structure, the system presents whatever was successfully parsed and flags the remainder for manual completion.

Exceptions:

- If the AI service is unreachable or times out, the system notifies the admin that generation failed and no draft is created.

Postconditions: A new draft flow exists, requiring explicit Tenant Admin review before it can be published; the AI Request is logged.

UC-04 — Receive and Route an Inbound Call

Actors: Caller, System (Telephony Layer, Execution Engine)

Preconditions: A DID is active and mapped to a deployed Flow Version.

Main Flow:

20. The caller dials the tenant's published phone number.
21. Asterisk receives the call and notifies the platform's telephony integration layer.
22. The system resolves the dialed DID to its tenant and deployed Flow Version.
23. The system creates a Call Session and begins executing the flow from its entry node.
24. The system plays the Greeting node's prompt.
25. The system presents a Menu node; the caller enters a DTMF digit.
26. The system follows the connection matching the caller's input toward a Queue node.
27. The caller is placed in queue and subsequently connected to an available agent.
28. The call ends; the system generates a CDR and emits analytics events for the full path taken.

Alternative Flow:

- 6a. If the caller enters no input before the timeout, the system replays the menu up to the configured retry limit.
- 6b. If the retry limit is exceeded, the system follows the node's configured fallback path (e.g., to voicemail).

Exceptions:

- 8a. If no agent becomes available within the queue's configured maximum wait time, the system triggers the queue's overflow action (e.g., route to voicemail).
- If the DID has no deployed flow, the system routes the call to a default fallback greeting and then disconnects.

Postconditions: The call is fully handled (connected to an agent, sent to voicemail, or ended), and a complete CDR exists for reporting.

UC-05 — Agent Handles a Transferred Call

Actors: Employee (Agent), Caller, System

Preconditions: The agent's SIP extension is registered and their status is Available.

Main Flow:

29. Flow execution reaches an Agent Transfer or Queue node and selects the agent per the routing strategy.
30. The system rings the agent's SIP extension.
31. The agent answers the call.
32. The system connects the caller and the agent, marks the agent's status as Busy, and updates the Call Session.
33. The call ends; the system updates the agent's status back to Available and finalizes the CDR.

Alternative Flow:

- 3a. If the agent does not answer within the configured ring timeout, the system tries the next agent per the routing strategy or triggers the queue's overflow action.

Exceptions:

- If the agent's extension is unregistered/unreachable at the time of routing, the system skips that agent and logs the event.

Postconditions: The call is connected to an available agent, or handled per overflow rules if none are available.

UC-06 — Caller Leaves a Voicemail

Actors: Caller, Employee, System

Preconditions: Flow execution has reached a Voicemail node with a configured recipient (employee or department).

Main Flow:

34. The system plays the configured voicemail greeting.
35. The caller records their message after the tone.
36. The caller ends the recording (via hangup or a configured stop key).
37. The system saves the Voicemail record, linked to the Call Session and recipient.
38. The system notifies the assigned employee/department of the new voicemail.

Alternative Flow:

- 2a. If the caller reaches the maximum recording duration, the system automatically stops recording and saves the message.

Exceptions:

- If storage of the recording fails, the system logs the error and, where possible, notifies the Tenant Admin of the failure.

Postconditions: A Voicemail record with status 'New' exists and the recipient has been notified.

UC-07 — View Reports and Analytics

Actors: Tenant Admin, Manager, Auditor

Preconditions: The user is authenticated and has reporting access for the requested scope.

Main Flow:

39. The user navigates to the Reporting dashboard.
40. The user selects a date range, and optionally a department or flow.
41. The system aggregates CDRs and Analytics Events matching the criteria, scoped to the user's tenant.
42. The system renders the dashboard (call volume, queue performance, flow drop-off, agent performance).
43. The user optionally exports the report as CSV or PDF.

Alternative Flow:

- 3a. If no data exists for the selected criteria, the system displays an empty-state message rather than an error.

Exceptions:

- If the user requests a scope outside their authorization (e.g., another department), the request is denied and logged.

Postconditions: The user has viewed and/or exported the requested report, without access to data outside their authorized scope.

UC-08 — Provision an Employee and SIP Extension

Actors: Tenant Admin

Preconditions: The tenant is active; the target department exists.

Main Flow:

44. The Tenant Admin creates a new Employee record and assigns it to a department.
45. The Tenant Admin provisions a new SIP extension for the employee.
46. The system generates SIP credentials and returns the provisioning details for the employee's device/softphone.
47. The Tenant Admin optionally assigns the employee to one or more Call Queues.

Alternative Flow:

- 2a. If the Tenant Admin instead links the employee to an existing unused extension, the system validates the extension is unassigned before linking.

Exceptions:

- If extension provisioning fails (e.g., naming conflict), the system displays the specific conflict and does not create a partial record.

Postconditions: The employee has an active SIP extension and, if assigned, is a member of the relevant queue(s).

UC-09 — Request AI-Generated Workflow Improvement Suggestions

Actors: Tenant Admin, AI Service (external)

Preconditions: A published flow with sufficient analytics history exists.

Main Flow:

48. The Tenant Admin opens an existing published flow's analytics view and selects 'Suggest Improvements'.
49. The system sends the flow structure and its aggregated analytics to the AI service.
50. The AI service returns a set of suggested changes with rationale (e.g., 'menu option 3 has a 40% drop-off; consider simplifying').
51. The system presents the suggestions to the Tenant Admin.
52. The Tenant Admin selects which suggestions to apply, generating a new draft version incorporating them.
53. The Tenant Admin reviews, validates, and publishes the new version (as in UC-02, steps 5-8).

Alternative Flow:

- 4a. The Tenant Admin may dismiss all suggestions without creating a draft.

Exceptions:

- If the AI service is unavailable, the system informs the admin that suggestions could not be generated at this time.

Postconditions: Either no change is made, or a new draft version incorporating selected AI suggestions is created for review.

UC-10 — Super Admin Suspends a Tenant

Actors: Super Admin

Preconditions: The target tenant exists in Active status.

Main Flow:

54. The Super Admin opens the platform administration console and selects the tenant.
55. The Super Admin chooses to suspend the tenant and provides a reason.
56. The system updates the tenant's status to Suspended and disables inbound call routing for all of its DIDs.
57. The system notifies the Tenant Admin of the suspension and the stated reason.
58. The system logs the action in the platform audit trail.

Alternative Flow:

- The Super Admin may later reactivate the tenant, restoring Active status and inbound routing.

Exceptions:

- If a call is in progress at the moment of suspension, the system allows that call to complete but blocks any new inbound calls.

Postconditions: The tenant is in Suspended status; its users cannot log in to perform configuration changes beyond viewing suspension status, and no new inbound calls are routed.

12. Acceptance Criteria

The platform will be considered acceptable for release/evaluation when the following criteria are demonstrably met:

- AC-01: A new tenant can complete registration, be approved by a Super Admin, and log in as a Tenant Admin without developer intervention.
- AC-02: A Tenant Admin can build, validate, publish, and deploy an IVR flow using only the visual builder, with no direct database or code changes.
- AC-03: An inbound test call to a deployed DID is correctly routed through the published flow, including at least one Menu, one Queue, and one Voicemail path.
- AC-04: A transferred call correctly connects to an available agent's SIP extension, and an unavailable agent correctly triggers the configured fallback.
- AC-05: A completed call produces an accurate CDR visible in the reporting dashboard within an acceptable delay (target: under 1 minute).
- AC-06: A Tenant Admin can generate a draft IVR flow via the AI assistant from a natural-language prompt, and the draft requires explicit publish action before going live.
- AC-07: Cross-tenant data access attempts (e.g., Tenant A user requesting Tenant B's report) are denied and logged.
- AC-08: Role-restricted actions (e.g., an Agent attempting to edit a flow) are denied at the API level, not only hidden in the UI.
- AC-09: Call recordings and voicemail are retrievable only by authorized roles and are retained/purged per the configured policy.
- AC-10: All non-functional requirements in Section 7 that are testable in the project's deployment environment (e.g., page response time, validation error surfacing) are verified through test execution.

13. Future Enhancements

- Multi-channel support (SMS, WhatsApp, web chat) reusing the existing flow-graph model.
- Outbound campaign management (reminders, surveys) built on the same node/flow concept.
- Flow simulation/test-call mode allowing a Tenant Admin to test a draft flow without a live phone line.
- Skills-based and language-based routing enhancements for queues.
- Callback-from-queue functionality.
- A/B testing between Flow Versions with comparative analytics.
- Tenant-facing public API and webhook system for external CRM/EHR integration.
- White-labeling and custom branding per tenant.
- Formal billing and usage-based metering module.
- Real-time live agent/queue monitoring dashboard.

14. Risks

Risk	Impact	Likelihood	Mitigation
Asterisk/PJSIP integration complexity underestimated	High	Medium	Prototype the ARI/AMI integration early; isolate it behind a clear internal contract.
External LLM API unavailability or rate limits	Medium	Medium	Treat AI features as asynchronous and optional; degrade gracefully if unavailable.
Cross-tenant data leakage due to missed scoping check	High	Medium	Enforce tenant scoping at the DAO layer with mandatory tenant-id parameters; add automated tests specifically for isolation.
Toll fraud / call flooding abuse	High	Low-Medium	Implement rate limiting, anomaly detection, and SIP credential hardening (NFR-SEC3, NFR-SEC4).
Absence of a full framework (Spring) leads to inconsistent layering over time	Medium	Medium	Enforce package-per-module structure and code review discipline (NFR-M1).
Recording/voicemail storage growth without retention enforcement	Medium	Medium	Implement configurable, enforced retention policies from the first release (FR-021, NFR-B2).
Scope creep given the breadth of an enterprise-style platform	Medium	High	Prioritize FRs (High/Medium/Low) and phase delivery; treat Section 13 items as explicitly out of initial scope.

15. Glossary

Term	Meaning
Agent	An employee who handles transferred or queued calls.
Business Hours Node	A node type that branches flow execution based on configured operating hours.
Draft	An editable, unpublished state of a Flow.
Entry Node	The single starting node of a Flow's graph.
Fallback Path	The connection followed when a node's primary outcome cannot be achieved (e.g., timeout, error).
Holiday Check Node	A node type that branches flow execution based on a configured holiday calendar.
Overflow	The configured behavior when a queue's maximum wait time is exceeded.
Retention Policy	Rules governing how long recordings, voicemail, or logs are kept before deletion.
Skills-based Routing	A queue strategy that routes callers to agents based on matched skill tags.
Tenant Isolation	The guarantee that one tenant cannot access another tenant's data or configuration.